

# Identifying Patterns of ASD Naturalistic Behavior Using Eye-Tracking Technology

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## INTRODUCTION

Autism Spectrum Disorder (ASD) affects social communication, interaction, and behavior. Early identification can support earlier intervention and better developmental outcomes. Traditional ASD assessment is mainly based on behavioral observation, while eye-tracking provides objective measurements of visual attention, including fixations, saccades, gaze dispersion, AOI transitions, and pupil dynamics.

### OBJECTIVE

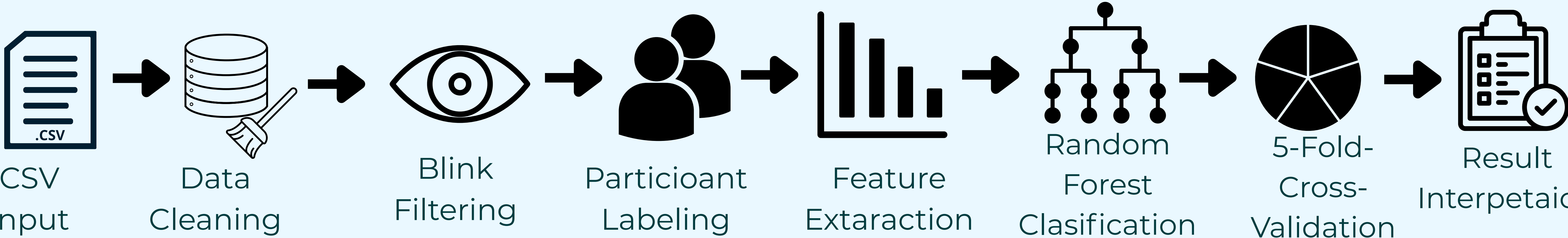
This project develops a Python-based research that processes eye-tracking CSV files, extracts interpretable gaze and pupil features, compares ASD and typically developing (TD) participants, and evaluates whether these features can support ASD-related pattern identification.

### RESEARCH QUESTION

Can ASD-related behavioral patterns be identified from eye movements and pupil dynamics during naturalistic visual exploration?

### PROPOSED SOLUTION

The system is an analytical pipeline that transforms raw eye-tracking recordings into structured behavioral features and evaluates their ability to distinguish between ASD and TD participants.



### FEATURE GROUPS

- Exploitation: Average fixation duration, maximum fixation duration, total AOI time.
- Exploration: Saccade frequency, average saccade length, AOI changes, gaze dispersion.
- Pupil Dynamics: Average pupil diameter, pupil range, pupil variability, pupil velocity, and pupil during fixation.

### TECHNOLOGIES

Python, Google Colab, pandas, NumPy, scikit-learn, Matplotlib, Seaborn, Tobii Pro Lab CSV exports, GitHub.

### DATASET

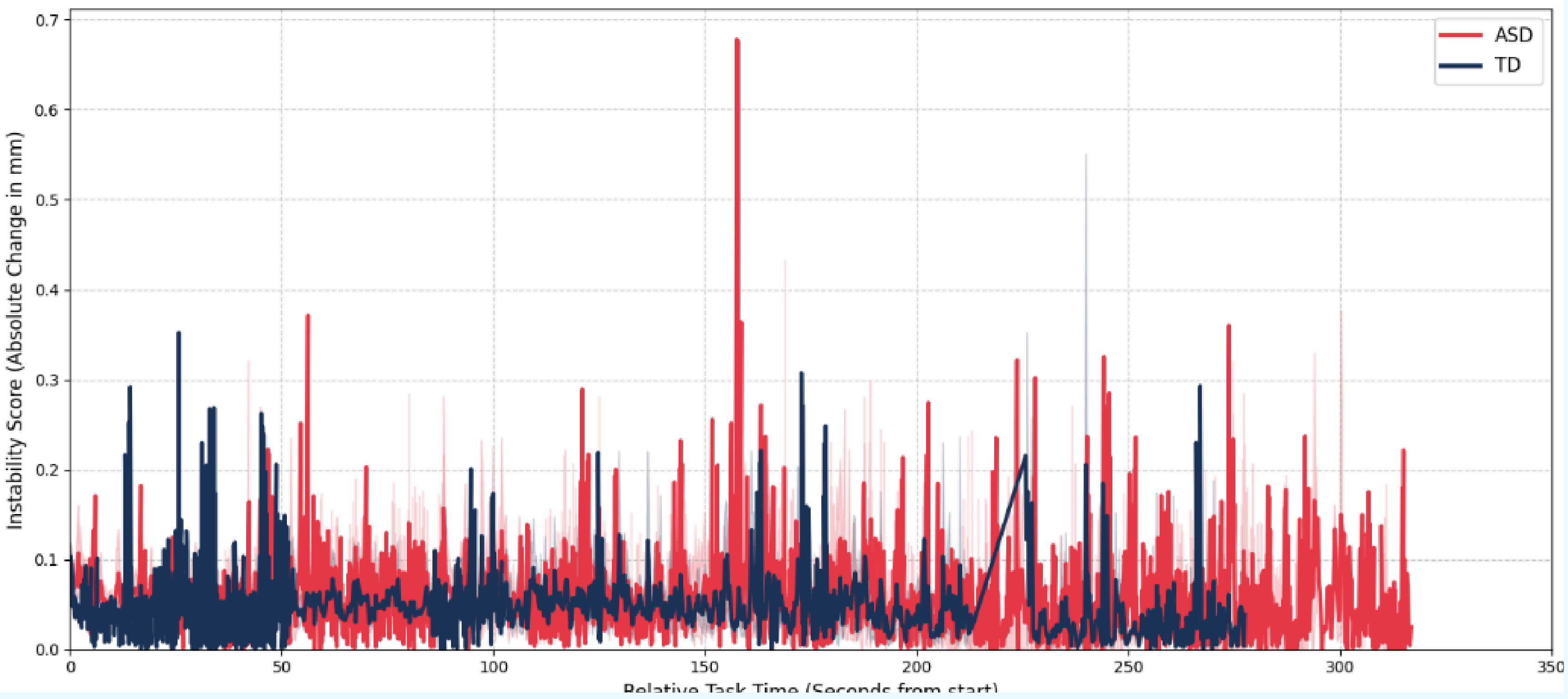
59 participants:  
30 TD  
29 ASD

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[ EXPLOITATION (Focus & Depth) ]
▼ Avg Fixation Duration: ASD generally had LOWER values (-22.5% compared to TD).
▼ Max Fixation Duration: ASD generally had LOWER values (-9.0% compared to TD).
▼ Total AOI Time: ASD generally had LOWER values (-39.2% compared to TD).

[ EXPLORATION (Scanning & Breadth) ]
▼ Saccade Frequency: ASD generally had LOWER values (-11.5% compared to TD).
▲ Avg Saccade Length: ASD generally had HIGHER values (+18.5% compared to TD).
▲ AOI Changes: ASD generally had HIGHER values (+10.7% compared to TD).
▲ Gaze Dispersion: ASD generally had HIGHER values (+24.2% compared to TD).

[ PUPIL DYNAMICS (Cognitive Load & Arousal) ]
▼ Avg Pupil: ASD generally had LOWER values (-35.1% compared to TD).
▼ Max Pupil: ASD generally had LOWER values (-31.1% compared to TD).
▼ Std Pupil: ASD generally had LOWER values (-43.5% compared to TD).
▼ Pupil CV: ASD generally had LOWER values (-42.7% compared to TD).
▼ Pupil Range: ASD generally had LOWER values (-39.3% compared to TD).
▼ Pupil Velocity: ASD generally had LOWER values (-60.0% compared to TD).
▼ Fixation Pupil: ASD generally had LOWER values (-35.5% compared to TD).
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Average Pupil Instability Over Time (Arousal Measure)  
Using 1-Second Window with 250ms Steps



=> Result: The ASD group shows HIGHER fluctuation frequency (higher arousal).  
Difference: 0.01435 mm

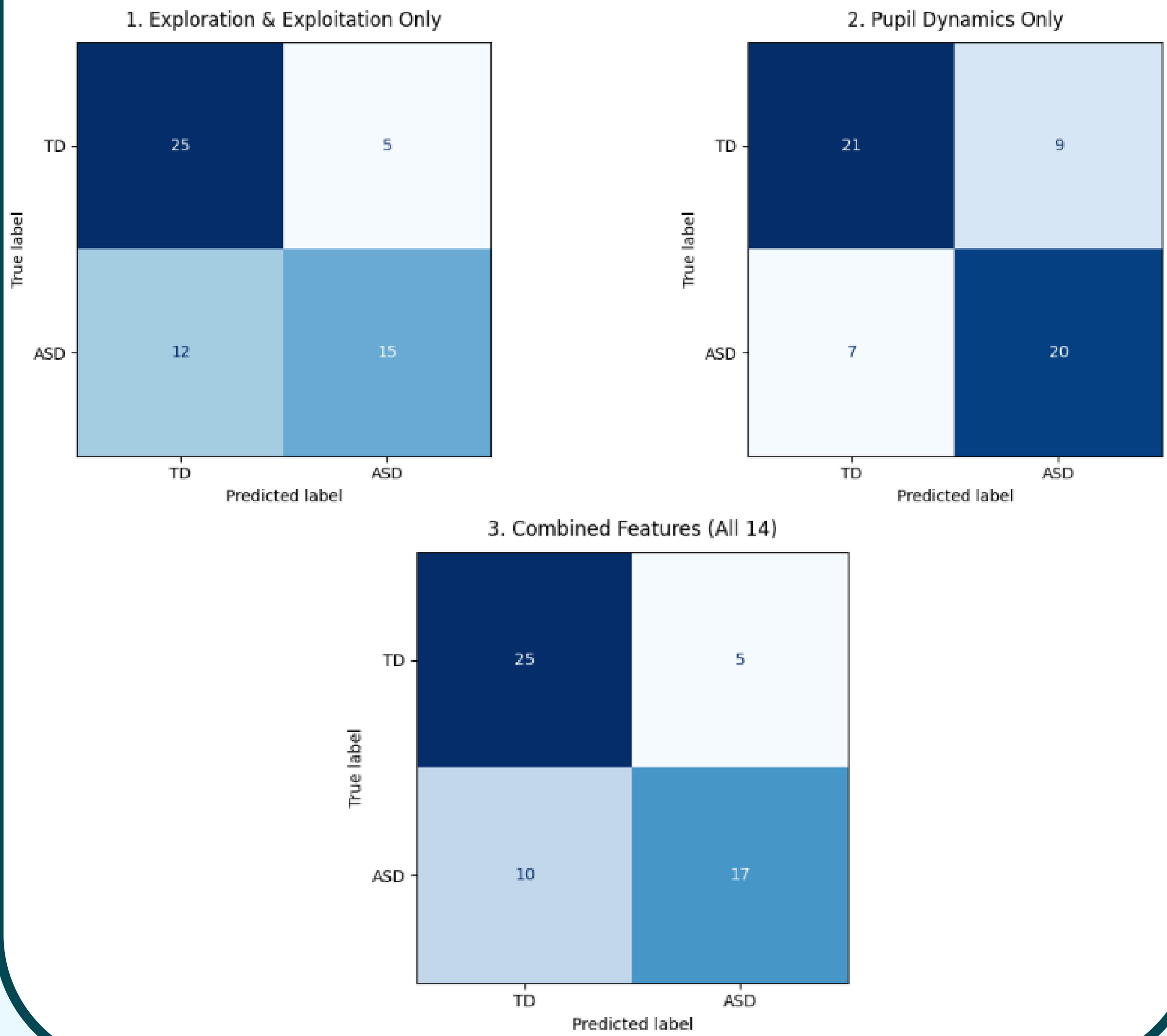
### Ablation Study Results:

- Exploration & Exploitation: 70% accuracy
- Pupil Dynamics: 72% accuracy
- Combined 14 Features: 74% accuracy

### Best Overall Result:

The combined feature set achieved the highest accuracy: 74%.

Key Insight: Pupil dynamics achieved the best ASD recall, correctly identifying 20 out of 27 ASD participants.



ASD Average Pupil Instability (Change per 250ms): 0.06349 mm  
TD Average Pupil Instability (Change per 250ms): 0.04915 mm